

2020 AMC 10B Problem 25 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 25. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 25

Problem:

Let $D(n)$ denote the number of ways of writing the positive integer n as a product

$$n = f_1 \cdot f_2 \cdots f_k$$

where $k \geq 1$, the f_i are integers strictly greater than 1, and the order in which the factors are listed matters (that is, two representations that differ only in the order of the factors are counted as distinct). For example, the number 6 can be written as 6, $2 \cdot 3$, and $3 \cdot 2$, so $D(6) = 3$. What is $D(96)$?

Answer Choices:

- A. 112
- B. 128
- C. 144
- D. 172
- E. 134

Solution:

Note that $96 = 2^5 \cdot 3$. Since there are at most six not necessarily distinct factors > 1 multiplying to 96, we have six cases: $k = 1, 2, \dots, 6$.

Now we look at each of the six cases.

- $k = 1$: We see that there is 1 way, merely 96.

- $k = 2$: This way, we have the 3 in one slot and 2 in another, and symmetry. The four other 2's leave us with 5 ways and symmetry doubles us so we have 10.
- $k = 3$: We have 3, 2, 2 as our baseline. We need to multiply by 2 in 3 places, and see that we can split the remaining three powers of 2 in a manner that is $3 - 0 - 0$, $2 - 1 - 0$ or $1 - 1 - 1$. A $3 - 0 - 0$ split has $6 + 3 = 9$ ways of happening ($24 - 2 - 2$ and symmetry; $2 - 3 - 16$ and symmetry), a $2 - 1 - 0$ split has $6 \cdot 3 = 18$ ways of happening (due to all being distinct) and a $1 - 1 - 1$ split has 3 ways of happening ($6 - 4 - 4$ and symmetry) so in this case we have $9 + 18 + 3 = 30$ ways.
- $k = 4$: We have 3, 2, 2, 2 as our baseline, and for the two other 2's, we have a $2 - 0 - 0 - 0$ or $1 - 1 - 0 - 0$ split. The former grants us $4 + 12 = 16$ ways ($12 - 2 - 2 - 2$ and symmetry and $3 - 8 - 2 - 2$ and symmetry) and the latter grants us also $12 + 12 = 24$ ways ($6 - 4 - 2 - 2$ and symmetry and $3 - 4 - 4 - 2$ and symmetry) for a total of $16 + 24 = 40$ ways.
- $k = 5$: We have 3, 2, 2, 2, 2 as our baseline and one place to put the last two: on another two or on the three. On the three gives us 5 ways due to symmetry and on another two gives us $5 \cdot 4 = 20$ ways due to symmetry. Thus, we have $5 + 20 = 25$ ways.
- $k = 6$: We have 3, 2, 2, 2, 2, 2 and symmetry and no more twos to multiply, so by symmetry, we have 6 ways.

Thus, adding, we have

$$1 + 10 + 30 + 40 + 25 + 6 = (A) \boxed{112}$$

OR

As before, note that $96 = 2^5 \cdot 3$, and we need to consider 6 different cases, one for each possible value of k , the number of factors in our factorization.

However, instead of looking at each individually, find a general form for the number of possible factorizations with k factors. First, the factorization needs to contain one factor that is itself a multiple of 3. There are k to choose from. That leaves $k - 1$ slots left to fill, each of which must contain at least one factor of 2. Once we have filled in a 2 to each of the remaining slots, we're left with $5 - (k - 1) = 6 - k$ twos.

Consider the remaining $6 - k$ factors of 2 left to assign to the k factors. Using stars and bars, the number of ways to do this is:

$$\binom{(6 - k) + k - 1}{6 - k} = \binom{5}{6 - k}$$

This makes $k \binom{5}{6-k}$ possibilities for each k . To obtain the total number of factorizations, add all possible values for k :

$$\sum_{k=1}^6 k \binom{5}{6-k} = 1 + 10 + 30 + 40 + 25 + 6 = (\text{A}) \boxed{112}$$

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